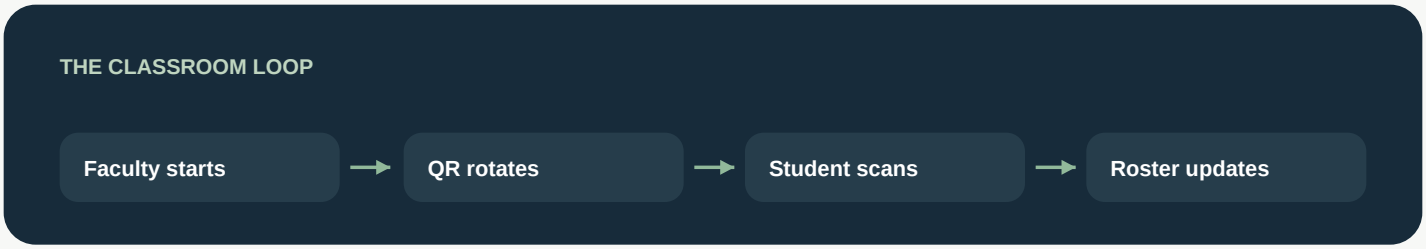
DIGITAL CLASS ATTENDANCE

Attendance, without the roll-call bottleneck.

SmartRoll helps faculty run a class check-in from a live QR code while students confirm attendance on their phones. No student app install is needed.



WHY IT FITS A CLASSROOM

Fast check-in

A changing QR and six-digit fallback code let students check in from the room display.

Clear ownership

Faculty sees a live roster; each student account links to an Enrollment ID.

Institution-ready

The planned access model separates institutions, memberships, records, and administrators.

DEPLOYMENT SHAPE

Cloudflare-hosted web app → Render API → Supabase PostgreSQL

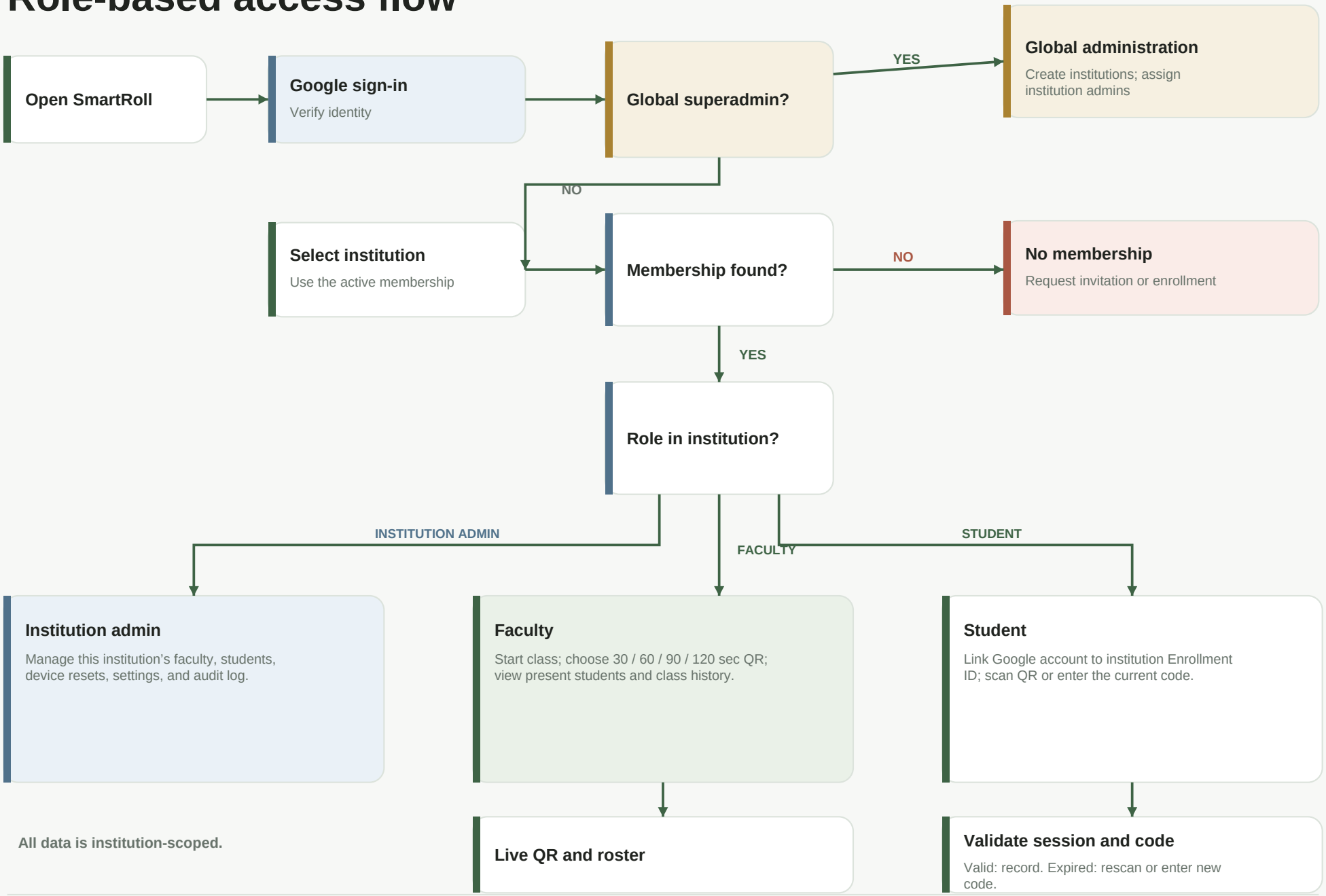
Google Identity Services verifies accounts; the API authorizes roles and writes attendance records.

SCOPE NOTE

Student and faculty workflows are in the current codebase. Global superadmin and institution-admin workflows are proposed next-phase work, not yet implemented.

TARGET MULTI-INSTITUTION MODEL

Role-based access flow



All data is institution-scoped.

FACULTY AND STUDENT JOURNEYS

How a class runs

A faculty member creates a named session. The display keeps the room focused on one QR, one current code, and a clear countdown.

FACULTY

1. Start a class

Enter a class name and choose a refresh interval: 30, 60, 90, or 120 seconds. Default: 60 seconds.

2. Project the code

The classroom display shows a QR link and six-digit code. Both refresh on the selected interval.

3. Follow attendance

The roster refreshes while class is active. Faculty can open a student's profile and review classes attended.

4. Close the session

Ending the session prevents further check-ins. Recorded attendance remains available in history.

STUDENT

1. Identify the account

Student signs in with Google and links the account to an Enrollment ID unique within the institution.

2. Open check-in

Scan the classroom QR with the phone camera or follow its check-in link. Sign-in returns to that class.

3. Confirm attendance

The API verifies student access, active session, and current six-digit code before saving attendance.

4. Recover or finish

Expired code? Scan again or enter the new code. Duplicate check-ins are acknowledged without adding another record.

ACCOUNT AND DEVICE

Google identifies the account. The first sign-in binds it to a browser-generated device ID; a device reset revokes its active API sessions. Sign-in tokens currently last 12 hours.

PRODUCT LIMIT

A rotating QR and device binding deter casual proxying; they cannot prove physical presence or stop a determined live relay.

PROPOSED ADMINISTRATION MODEL

A platform for many institutions

The multi-institution layer is the proposed next phase. It adds scoped administration around the existing attendance workflow.

PLATFORM-WIDE

Global superadmin

You are the platform owner. Create and suspend institutions, assign institution admins, see global health and activity, and review the global audit trail.

TENANT-SCOPED

Institution admin

Manage faculty and students only in the selected institution. Reset student devices, review attendance, and set local enrollment and QR policies.

DATA AND ACCESS RULES

Membership-scoped roles	A person’s role is attached to their institution membership, not globally to their Google account.
Institution-local IDs	Enrollment IDs are unique within an institution; the same ID can exist at another institution.
Tenant-isolated records	Sessions, attendance, device controls, and reports are always filtered by institution.
Traceable admin actions	Role changes, device resets, exports, and settings changes write actor, time, target, and institution to an audit log.
Safe removal	Removing faculty access revokes sign-in and access but preserves historical class and attendance records.

CURRENT CODE VS. PROPOSED

Google student/faculty sign-in, sessions, QR check-in, attendance, student profiles, and device reset exist in the codebase. Global superadmin, institution setup, and tenant-scoped institution admin are not implemented yet.

PITCH SUMMARY

Product foundation and next steps

CURRENT APPLICATION

Web client

React + Vite frontend hosted on Cloudflare. Works in a modern mobile or desktop browser.

Attendance API

FastAPI on Render verifies Google tokens, enforces roles, generates codes, and records attendance.

Data store

Supabase PostgreSQL stores accounts, device registrations, class sessions, and attendance.

DELIVERY ROADMAP

01 **Institution foundation**

Add institutions, memberships, institution-scoped roles, and tenant-safe data access.

02 **Admin operations**

Build global institution setup, institution faculty/student controls, device reset, and audit log.

03 **Operational reporting**

Add institution-level attendance exports, activity summaries, and configurable retention policies.

THE PITCH

SmartRoll turns attendance into a quick, visible classroom workflow. Faculty controls the session; students confirm with a rotating QR; institution-level administration can scale the model while keeping each school's records separate.